

# Semiorder

Model — ai; produced by Claude Fable 5.1 (Cowork session).

## 0.1 Package

- **Completeness.** For all  $p, q \in \Delta(X)$ :  $p \succcurlyeq q$  or  $q \succcurlyeq p$ .
- **Quasi-transitivity.** Strict preference is transitive: if  $p \succ q$  and  $q \succ r$  then  $p \succ r$ .
- **Closed-graph continuity.** The set  $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$  is closed in the product topology.
- **Transitivity.** For all  $p, q, r \in \Delta(X)$ : if  $p \succcurlyeq q$  and  $q \succcurlyeq r$  then  $p \succcurlyeq r$ .
- **Independence.** For all  $p, q, r \in \Delta(X)$  and  $\lambda \in (0,1]$ :  $p \succcurlyeq q$  if and only if  $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$ .

## 0.2 Construction

A semiorder with a just-noticeable difference. Fix  $u : X \rightarrow \mathbb{R}$  non-constant and  $\varepsilon > 0$ , and let  $p \succcurlyeq q$  iff  $EU_u(p) \geq EU_u(q) - \varepsilon$ . Completeness: one of the two differences is  $\geq 0 > -\varepsilon$ . Quasi-transitivity:  $p \succ q$  iff  $EU_u(p) > EU_u(q) + \varepsilon$ , and this is transitive. Closed graph: the graph is the preimage of  $[-\varepsilon, \infty)$  under a continuous map. Transitivity fails: pick  $p, q, r$  with  $EU_u$  equal to  $0, -0.8\varepsilon, -1.6\varepsilon$ ; then  $p \succcurlyeq q \succcurlyeq r$  but not  $p \succcurlyeq r$ .

## 0.3 Notes

Sanity-checked numerically in `checks/countermodels.py`. The relation is Luce's semiorder; the verification that it fits this framework is AI-produced and unchecked.

## 0.4 Sources

- Luce (1956), Semiorders and a theory of utility discrimination, *Econometrica* 24

Record: `topics/decision-theory/models/semiorder.yaml`